

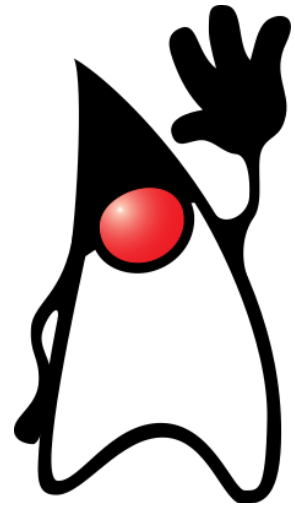
Java SE and JVM Fundamentals

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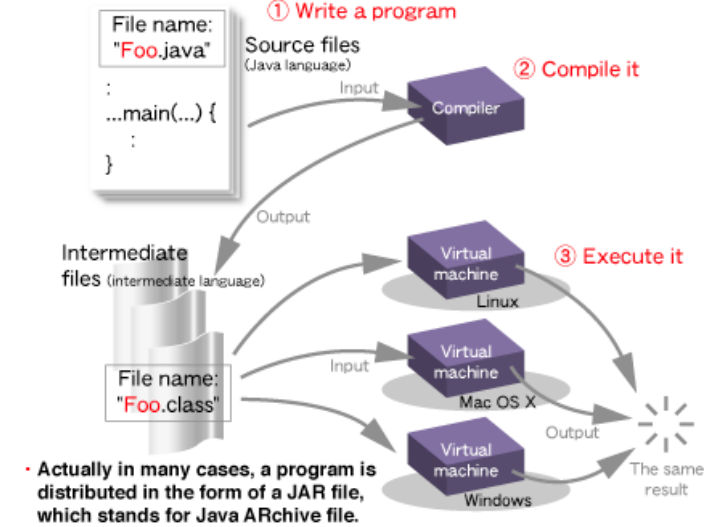
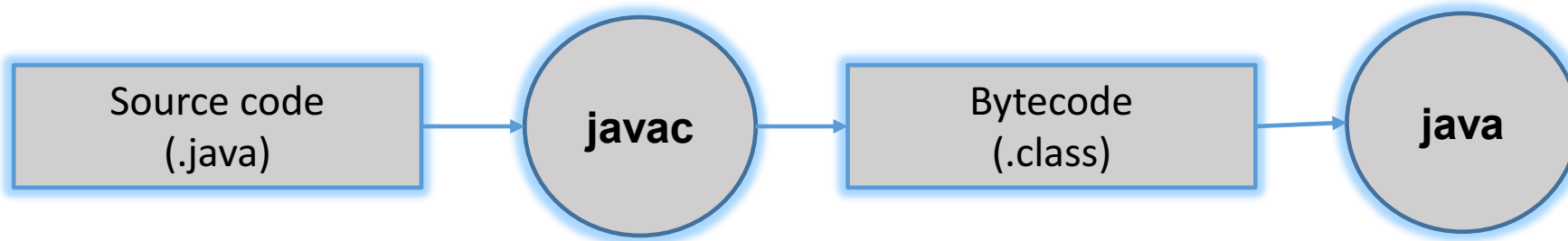
What is Java?



- 1991 – 1995 - it was first developed by **James Gosling** at Sun Microsystems (2010 - Oracle Corporation)
 - Oak, Green Project
- Java is a **general-purpose, concurrent, strongly typed, object-oriented** programming language
- Java can be embedded in web pages
- Write once, run anywhere (platform independent)
 - compiled Java code can run on many different operating systems.
- Java LTS versions like **JDK 21** and **JDK 25** are available under Oracle's No-Fee Terms and Conditions (NFTC) license, which allows free use for most purposes, including production, for users and commercial use.
 - **26 (latest release)**
- 2006, Sun announced OpenJDK as FOSS (Free and Open-Source Software) under the terms of the GNU General Public License (GPL). The OpenJDK project itself launched publicly in 2007.
- Java is a *de facto* standard, controlled through the Java Community Process.



Java Compilation and Execution



- A Java program isn't a single executable file, but instead, it is composed of many individual .class files, each of which corresponds to a single Java class.
- **Java programs are portable.** They can be run on any platform without being recompiled.
- Compilers of most of the programming languages, translate the source code from a high-level programming language directly to a machine code of a given platform, to create an executable program
 - Because different platforms have different machine codes, compilers are different also.
- The Java compiler compiles the source code into the platform-independent bytecode, which is the instruction set of an abstract computing machine, called the **Java Virtual Machine - JVM**.
 - Java bytecode is an intermediate code, because it is not understandable and executable by the real computing device
 - **JVM is platform-dependent**